

Linux Quick Reference Guide

Essential commands for developers and DevOps

Covers: navigation, files, permissions, processes,
SSH, Git, networking, and PowerShell equivalents

1. Getting Started

Linux is a Unix-like operating system widely used on servers, cloud VMs, and developer machines. Most commands run in a shell (bash is the default). Paths use forward slashes (/). Home directory is ~ (e.g. /home/username).

Open a terminal

```
Ctrl + Alt + T      # Ubuntu / most desktops
ssh user@hostname    # Connect to remote Linux server
```

Essential shortcuts

Shortcut	Action
Tab	Auto-complete file/command names
Ctrl + C	Cancel running command
Ctrl + D	Exit shell / EOF
Ctrl + L	Clear screen (or: clear)
Up / Down	Browse command history
man <cmd>	Show manual page for command

2. File System Navigation

```
pwd          # Print current directory
ls           # List files
ls -la       # List all files with details
cd /path/to/dir # Change directory
cd ~         # Go to home directory
cd ..        # Go up one level
mkdir myfolder # Create directory
mkdir -p a/b/c # Create nested directories
```

Common directories:

Path	Purpose
/home/user	User home directory (~)
/etc	System configuration files
/var/log	Log files
/tmp	Temporary files
/usr/bin	User-installed programs
~/.ssh	SSH keys and config

3. File Operations

```
touch file.txt      # Create empty file
cp source dest      # Copy file
cp -r src/ dest/    # Copy directory recursively
mv old new          # Move or rename
```

```
rm file.txt           # Delete file
rm -rf folder/        # Delete directory (careful!)
cat file.txt          # View file contents
less file.txt         # View file (scrollable, q to quit)
head -n 20 file.txt   # First 20 lines
tail -n 20 file.txt   # Last 20 lines
tail -f /var/log/syslog # Follow log in real time
find . -name '*.log'  # Find files by name
grep 'error' file.log # Search text inside files
```

4. File Permissions

Each file has read (r), write (w), and execute (x) permissions for owner, group, and others. Example: -rw-r--r-- means owner can read/write, group and others can read only.

```
ls -la              # Show permissions
chmod 755 script.sh # rwxr-xr-x
chmod +x script.sh  # Make executable
chown user:group file # Change owner (needs sudo)
sudo command        # Run as administrator (root)
```

Numeric	Symbolic	Meaning
7	rwx	Read + Write + Execute
6	rw-	Read + Write
5	r-x	Read + Execute
4	r--	Read only

5. Process Management

```
ps aux              # List all running processes
top                 # Interactive process viewer
htop                # Better top (if installed)
kill PID            # Terminate process by ID
kill -9 PID         # Force kill
pkill nginx         # Kill by process name
nohup cmd &         # Run in background, survives logout
jobs                # List background jobs
fg %1               # Bring job 1 to foreground
```

6. Package Management

Debian / Ubuntu (apt)

```
sudo apt update     # Refresh package list
sudo apt upgrade    # Upgrade installed packages
sudo apt install nginx # Install a package
sudo apt remove nginx # Remove a package
apt search keyword  # Search packages
```

RHEL / CentOS / Fedora (yum/dnf)

```
sudo dnf install nginx # Install package
sudo dnf update        # Update all packages
sudo dnf remove nginx  # Remove package
```

7. SSH & Remote Access

SSH (Secure Shell) lets you securely connect to remote servers and authenticate using passwords or key pairs.

Connect to a server

```
ssh user@192.168.1.10 # Connect by IP
ssh user@hostname.com # Connect by hostname
ssh -p 2222 user@host # Custom port
```

Generate SSH key pair

```
ssh-keygen -t ed25519 -C 'your_email@example.com'
                # Creates ~/.ssh/id_ed25519 (private)
                # and ~/.ssh/id_ed25519.pub (public)

cat ~/.ssh/id_ed25519.pub # Display public key to copy
ls -la ~/.ssh             # List SSH directory
```

Add key to GitHub / GitLab

Copy the entire single line from the .pub file (starts with ssh-ed25519 or ssh-rsa). Paste it in GitHub Settings > SSH and GPG keys, or GitLab User Settings > SSH Keys. The same public key can be used on multiple services.

Test SSH connection

```
ssh -T git@github.com # Test GitHub
ssh -T git@gitlab.com # Test GitLab
```

Copy files over SSH (SCP)

```
scp file.txt user@host:/remote/path/
scp -r folder/ user@host:/remote/path/
scp user@host:/remote/file.txt ./local/
```

8. Git Essentials

```
git clone git@github.com:user/repo.git
git status # Show changed files
git add file.txt # Stage a file
git add . # Stage all changes
git commit -m 'message' # Commit staged changes
git push # Push to remote
git pull # Fetch and merge remote changes
git branch # List branches
git checkout -b feature # Create and switch branch
git log --oneline # Compact commit history
git diff # Show unstaged changes
```

9. Text Processing & Pipes

The pipe (|) sends output of one command as input to another. Redirection: > writes to file, >> appends to file.

```
cat file | grep error    # Filter lines containing 'error'
cat file | wc -l         # Count lines
sort file.txt           # Sort lines alphabetically
uniq file.txt           # Remove adjacent duplicates
cut -d',' -f1 data.csv  # Extract first CSV column
sed 's/old/new/g' file  # Replace text
echo 'hello' > out.txt   # Write to file (overwrite)
echo 'more' >> out.txt   # Append to file
```

10. Networking

```
ip addr                 # Show IP addresses
ip route                # Show routing table
ping google.com         # Test connectivity
curl https://api.com    # HTTP request (GET)
curl -O https://url/file.tar.gz # Download file
wget https://url/file   # Download file
netstat -tulnp          # Show listening ports
ss -tulnp               # Modern netstat alternative
nslookup google.com     # DNS lookup
dig google.com           # Detailed DNS lookup
```

11. System Information

```
uname -a               # Kernel and system info
hostname               # Machine name
whoami                 # Current username
id                     # User and group IDs
df -h                  # Disk space usage
du -sh folder/         # Directory size
free -h                # Memory usage
uptime                 # System uptime and load
lscpu                  # CPU information
systemctl status nginx # Service status (systemd)
sudo systemctl start nginx # Start service
sudo systemctl enable nginx # Start on boot
```

12. PowerShell vs Linux Cheat Sheet

If you use Windows PowerShell, these are the Linux/bash equivalents for common tasks.

PowerShell (Windows)	Linux (bash)
Get-ChildItem	ls / ls -la
Get-Content file	cat file
Set-Content file	echo 'text' > file
Copy-Item	cp
Move-Item	mv
Remove-Item	rm
New-Item -ItemType Directory	mkdir
Select-String 'pattern'	grep 'pattern'
\$env:USERPROFILE	~ or \$HOME
\$env:PATH	\$PATH
Clear-Host	clear
Get-Process	ps aux
Stop-Process	kill
Test-Connection host	ping host
Invoke-WebRequest	curl / wget

13. Tips & Best Practices

- Always double-check before running `rm -rf`
- Use tab completion to avoid typos
- Read man pages: `man ls`, `man grep`, etc.
- Never share your private SSH key (`~/.ssh/id_*` without `.pub`)
- Use `sudo` only when needed
- Keep systems updated: `apt update && apt upgrade`
- Use `ssh-keygen -t ed25519` for new SSH keys (more secure, shorter)
- Set up `~/.ssh/config` for easier SSH connections
- Use `| less` for long output: `ls -la | less`

Example `~/.ssh/config`

```
Host myserver
  HostName 192.168.1.10
  User ubuntu
  IdentityFile ~/.ssh/id_ed25519
  Port 22
```

```
# Then connect with: ssh myserver
```